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## Chapter 1 – Introduction

Nowadays' increasing frequency and severity of natural disasters has highlighted the effects of climate change and global interconnectedness, exposing critical gaps in traditional disaster impact assessment methods. Conventional measures often rely on reactive measures which struggle to address evolving patterns and complex socioeconomic vulnerabilities. This thesis investigates the potential of Machine Learning (ML) to transform disaster economics by enhancing predictive accuracy while critically examining the methodologies, data quality and ethical challenges inherent to such applications.

Existing studies have explored ML applications in hazard prediction and response coordination, but few have addressed the comprehensive integration of technical innovation with ethical governance frameworks. This work aims to fill a critical gap by focusing on the Emergency Events Database (EM-DAT) to develop predictive models for economic damage estimation while establishing guidelines for responsible AI deployment under the European Union's Artificial Intelligence Act.

The primary objective is threefold: first, evaluating the performance of ML models in predicting disaster economic impact using EM-DAT; second, analyzing system biases and data limitation potentially affecting model reliability and third, proposing an ethical framework for AI integration in disaster management following the Conformity Assessment Procedure for AI (capAI). The thesis begins with a theoretical foundation in disaster management frameworks and AI applications (Chapter 2), followed by an empirical model comparison (Chapter 3) and an ethical evaluation using the capAI framework (Chapter 4), outlaying ML's technical potential while mitigating risks in high-stakes humanitarian contexts.

# Chapter 2 – Disaster Management Theory

## 2.1 What is Disaster Management

Disaster management is a comprehensive and coordinated approach that integrates all activities addressing the impacts of disasters, necessary to sustain a community's ability to prepare for, protect against, respond to and recover from both natural and human-induces disasters. It is a multi-jurisdictional, multi-sectoral, and multi-disciplinary activity that requires collaboration among national and local governments, civil society organizations and private sector to achieve common goals (NDMF, 2010).

Effective disaster management recognizes that recovery is not a linear process but rather a cyclical interdependent one, where phases as such may be defined by boundaries determined by the event itself and the occurrence of simultaneous events. In fact, disasters usually occur in succession and can overlap, as noticeable in the EM-DAT dataset that will later be analyzed. Such complex and dynamic approach ensures that recovery and risk reduction are community-driven and locally defined, implying the contribution of local players in support of the identification of recovery needs and priorities. The national government, or the federal as in the case of the United States with the National Disaster Recovery Framework (NDRF), supports these efforts by providing assistance and resources (FEMA, 2024).

Disaster management also involves the integration of Disaster Risk Reduction (DRR) in its broad action planning. As described in the Disaster Risk Management (DRM) Systems Analysis Guide, it is crucial to assess existing structures, identify gaps and design community-based strategies that address the root causes of vulnerability in disaster-prone areas and

high-risk populations (FAO, 2008). This complete overview on disaster management allows more accurate displacement of response and recovery resources, considering both actual and aggravating circumstances of each instance.

The disaster management framework cycle typically includes four phases.

Prevention and mitigation: Includes actions taken to reduce the impact of future disasters, spanning from technical ones as risk assessment, hazard identification, implementation of preventive measures; to social ones as education, land use planning and relationship building among key stakeholders.

Preparedness: This phase includes planning resource allocation, training personnel, develop early warning systems, evacuation plans and emergency protocols.

Response: Actions taken during and immediately after a disaster aiming to save lives, protect infrastructures and safeguard basic human needs. This phase includes the participation of local agencies and volunteers, first responders and humanitarian assistance.

Recovery: Efforts to restore the health, social, economic and environmental systems of affected communities, with an emphasis on resilience. Recovery planning should begin during the response phase and continue long after the immediate crisis.

A successful disaster management system is characterized by collaboration across government agencies, nongovernmental organizations, the private sector and the whole community. Increasingly relying on data-driven decision-making, appropriate recovery planning should rely on continuous assessment and fostering future resilience. Over the past few decades, the

enormous increase in computational capacity and speed of available data collection tools has allowed to collect an enormous and higher-quality amount of data.

### 2.2 Machine Learning in Disaster Management

The introduction of Machine Learning (ML) into disaster management has revolutionized our society's capabilities of predicting, preparing for, responding to and recovering from catastrophic events. Leveraging datasets from Internet of Things devices, satellite imagery and historical records, ML algorithms enable stakeholders such as those mentioned in the Section 2.1 to optimize resource allocation through the analysis of complex patterns. Techniques used span from categorization and classification of satellite images to building inspection records on social media posts. Notably, such flexibility of ML has enabled its application across the disaster management cycle.

Mitigation focuses on minimizing disaster impact risk before crises occur, and ML models excel in identifying risk factors by analyzing historical data about infrastructure conditions and environmental changes. A study by Shukla and Kumar, 2022, highlights ML's results in predicting seismic vulnerability in urban areas by analyzing building materials and topographic data. The World Bank's case study on Guatemala City demonstrates the ability of Convolutional Neural Networks (CNNs) in processing street-view imagery to detect "soft-story" buildings prone to collapse in case of earthquake, achieving 85% accuracy (World Bank, 2018).

Preparedness consequently relies on accurate early warning and development of evacuation plans. With this scope, ML has proven its ability in processing real-time data from weather sensors, social media and satellite; with notable applications in flood prediction, where algorithms

combine radar imagery and Digital Evolution Models (DEMs) to simulate water flow patterns and issue alerts (World Bank, 2018). Similarly, another application was found in simulating traffic dynamics during hurricanes, reducing bottlenecks and saving critical response time (Shukla et Kumar, 2022). Such systems have high potential in democratizing risk communication, ensuring vulnerable populations can receive timely alerts through mobiles.

During disasters, ML can accelerate response by automating damage detection to coordinate relief efforts. Indeed, imagery from Unmanned Aerial Vehicle collected in Vanuatu and crowdsourced data has been put into CNNs to classify cyclone damage levels with 63% accuracy, enabling rapid search and rescue operations (World Bank, 2018). Other applications include Natural Language Processing (NLPs) tools able to analyze social media posts to map distress signals and efficiently allocate resources; and Random Forest algorithms deployed to improve distribution of medical kits and shelter materials depending on predicted demand (Shukla et Kumar, 2022).

Post-disaster recovery benefits from ML's ability to analyze socio-economic vulnerabilities and guide reconstruction. A notable application is a mapping project carried out in Sri-Lanka, which used satellite imagery and census data to train regression models which have proven able to explain 61% of poverty variance, informing equitable aid distribution (World Bank, 2018). ML-driven platforms have also been used in assessing repair costs with 3D building scans, to allocate funds depending on damage (Shukla et Kumar, 2022).

Despite its proven potential, adoption of ML systems faces hurdles due to its own nature. Biased data used for training can perpetuate inequalities,

and models trained on high-income urban areas may fail to replicate performance in low-income rural settings. A study carried out by the World Bank in 2018 highlights the need for geographically diverse datasets to avoid pitfalls, citing the study in Guatemala City where local street-view imagery improved model accuracy. Additionally, many algorithms may come across as “black boxes”, a term that indicates systems whose input and output is clear, but users are unaware of the processes in between (IBM, 2024). Such issue may erode public trust and highlights the urgency of Explainable Artificial Intelligence (XAI) in contest of social applications, to give human users full accessibility to the systems and guarantee comprehension (IBM, 2023). The World Bank advocates for open-source platforms like TensorFlow and collaborative “trainathons” to crowdsource labeled datasets, as demonstrated in the OpenStreetMap initiatives (World Bank, 2018).

### 2.3 The EM-DAT Database

The Emergency Events Database (EM-DAT) is a globally recognized repository of disaster data, maintained by the Centre for Research on the Epidemiology of Disasters (CRED) at the University of Louvain in Belgium. Established in 1988 originally in collaboration with the World Health Organization (WHO), it was conceived as part of a broader global emergency management data integration system aimed at supporting disaster preparedness and humanitarian aid. The EM-DAT includes more than 26,000 records of natural and technological disasters from 1900 to the present, providing critical insights into their human and economic impacts. Natural hazards follow the 2014 IRDR Peril Classification and Hazard Glossary with up to four additional classification levels, while technological hazards are categorized into three main types: transport, industrial and miscellaneous accidents.

Since its creation, the EM-DAT has supported evidence-based disaster preparedness, response and risk-reduction. It aggregates data from various sources, including but not limited to United Nations agencies, governmental reports, reinsurance companies (as Munich Re) and media outlets (as ReliefWeb). Disasters included in the dataset must meet at least one among the following criteria: 10 or more deaths, 100 or more people affected, declaration of a state of emergency or international aid appeal (EM-DAT Documentation: Specific Biases, n.d.). As of 2024, the EM-DAT dataset contains over 26,000 entries, covering more than 120 hazard types categorized into natural, technological and complex disasters. Being an open-access platform, it has over 92,000 users registered, predominantly from academia and non-governmental organizations.

In 2023 EM-DAT adopted FAIR (Findable, Accessible, Interoperable, Reusable) Guiding Principles for scientific data management and stewardship, which emphasize machine-actionability in response to humans' increasing reliance on computational support due to the increase of volume and complexity of data (GO FAIR, n.d.). Key changes included the adoption of a 4-tier hazard classification aligned with UN standards, UN M49 country and region naming conventions and a public documentation portal detailing methodology. This shift has increased transparency but has exposed historical data inconsistencies through labeling pre-2000 entries as "historical" due to collection practices (EM-DAT Public Table, n.d. and Kripa et al., 2022).

### 2.3.1 Data Content and Structure

The EM-DAT uses a relational model with three core components, namely disaster events identified with a unique 8-digit ID (DisNo.) combining year

and sequential code; instance-specific human and economic loss data and validated references for each metric.

In more detail, the Human Impact variables include Total Deaths, accounting also for missing persons presumed dead, and Total Affected, calculated as the sum of No. Injured, No. Affected and No. Homeless. It includes multipliers, x5 for developing nations and x3 for industrialized, when only housing data is accessible. The Economic Impact variables instead include Insured Damages, representing infrastructure losses, and Reconstruction Costs, both rarely reported (EM-DAT Specific Biases and Public Table, n.d. and Kripa et al., 2022).

### 2.3.2 Limitations

Despite its comprehensive approach, EM-DAT faces several inherent limitations that affect data quality and completeness.

As of documentation, three main types of data quality issues can be identified. First, Missing Disaster Events that meet the inclusion criteria but are not recorded in EM-DAT, as highlighted by cross-comparison studies. Second, disasters existing within the database, but having incomplete information regarding impact variables, especially economic loss data. Lastly, some well-documented disasters have attributes that differ from other sources or contain inaccuracies (EM-DAT Documentation: General Issues, n.d.). The completeness of the dataset inevitably reflects the coverage of its sources, which has improved over the last 30-40 years thanks to enhanced reporting practices. However, the absence of universally applied protocols for reporting disaster impacts and losses leads to inconsistencies within the dataset with respect to other sources.

EM-DAT's inclusion thresholds create the potential for bias, were disasters just below one of the rules remain undocumented despite potentially

contributing to significant cumulative impact. This could occur in cases of connected events spread over time or disasters in regions with poor reporting infrastructures.

As of the second category, namely recorded disasters with incomplete information, an analysis of records between 1990 and 2020 has shown that Reconstruction Costs are missing in 96.2% of disasters, Insured Damages in 88.1% and Total Estimated Damages in 41.5%. On the other side, human impact variables show better reporting, with missing data proportions ranging from 1.3% to 22.3% (Jones et al.,2022). Such disparity comes from the fact that traditional aid allocation models rely on human impact figures, creating stronger incentives for their reporting.

Logistic regression analysis showed that missing data in EM-DAT is not random but systematically associated with factors including year the disaster occurred, income-group classification of affected country, disaster severity and type. For total estimated damages, disaster occurred after 2002 and in lower-income countries were more likely to have missing data, as were disasters with prolonged effects like droughts, epidemics and extreme temperatures events (Jones et al.,2022).

### 2.3.3 Biases

Beyond general limitations, EM-DAT exhibits several systematic biases that require careful consideration when using the database for research or policy decisions, reflecting broader challenges in disaster data collection worldwide.

Time biases result from unequal reporting quality and coverage over different historical periods (Gall et al., 2009). The significant increase in disaster records after the 1960s coincides with the creation of the Office of US Foreign Disaster Assistance (OFDA) and the subsequent establishment

of CRED in 1973 (CRED, 2023). The advent of personal computers and satellites in 1970s and the World Wide Web in 1990s further enhanced reporting capabilities.

Different types of hazards receive unequal reporting and coverage, leading to systematic disparities within the database (Gall et al., 2009). Biological hazards such as epidemics are particularly underreported. Similarly, research on heatwaves illustrate this bias, showing significant underreporting in regions like Sub-Saharian Africa despite evidence of health impacts from extreme heath events (Brimicombe et al., 2021). The combination of inadequate monitoring systems and higher reporting thresholds for certain disaster types creates systematic gaps in the database (EM-DAT Documentation: Specific Biases, n.d.).

Indeed, high-impact disasters attract more media attention and reporting resources, resulting in better documentation compared to low-impact events. EM-DAT's entry criteria inevitably introduce a form of threshold bias by excluding events below certain impact level. This bias particularly affects understanding of chronic, slow-onset or recurrent disasters that may not individually catch the attention of media outlets but create significant cumulative burden over time (Marulanda et al., 2010).

Spatial inconsistencies in data coverage create geographic biases, with regions like Sub-Saharan Africa often underrepresented (Osuteye et al., 2017 and Harrington et al., 2020). These disparities often stem from differences in investments in national disaster loss monitoring systems, international media coverage of different regions, language barriers in reporting and documentations, and geopolitical factors affecting humanitarian attention and resource allocation. Countries with well-developed disaster monitoring infrastructures and significant international

media presence are likely to have more complete representation in EM-DAT, leading to geographic bias.

The dataset exhibits inconsistent reporting across different types of impacts. Mortality data shows the highest completion rates, with only 1.3% missing for Total Deaths, while economic impacts have substantially higher rates of missing data. This creates systematic imbalances in understanding different dimensions of disaster impact. Additionally, less tangible and less easily accountable losses tend to be underestimated or completely overlooked. Indirect and long-term impacts such as disease outbreaks, mental health consequences or educational disruptions following disasters are rarely captured systematically. Similarly, economic reporting favors physical damages and insured losses over indirect impacts or losses in informal economies (Gall et al., 2009).

In conclusion, missing data patterns in EM-DAT are associated with specific factors, creating the potential for biases in analysis that do not account for these mechanisms. Jones et al., 2022, suggested that missing data follows patterns consistent with Missing At Random (MAR) or potentially Missing Not At Random (MNAR), rather than Missing Completely At Random (MCAR). When data are MAR, their absence is systematically related to the observed but not to the unobserved data; while when data are MNAR the link is with unobserved data not measured by the research. This distinction is crucial for interpretation and analysis, as simple deletion methods could produce bias when data is not MCAR, which would be the case if the pattern was independent of both observable and unobservable data.

In summary, the study by Jones et al. revealed that economic data was more likely to be missing in lower-income countries, human impact data showed better reporting for disasters in lower-income countries, probably

because of higher humanitarian aid requirements, both economic and human impact data showed higher rate of missing values for records after 2002, and certain disaster types, such as droughts, epidemics and extreme temperatures, showed systematically higher rates of missing economic data.

### 2.3.4 Addressing Limitations and Biases

In September 2023 EM-DAT significantly expanded its documentation, creating a comprehensive 15,000-word resource that explicitly addresses known limitations and biases, helping users make informed decisions about data interpretation and analysis (CRED Crunch, 2024). Pre-2000 records are flagged in a "Historic" field, as they should be considered of less quality due to time bias issues (EM-DAT Documentation: EM-DAT Public Table, n.d.). A study by Jones et al. recommends excluding pre-2000 data from trend analyses to avoid confusing better reporting with changes in disaster patterns. The EM-DAT remains committed to ongoing improvements in data quality and coverage, as visible through recent initiatives such as the implementation of a versioning system for database updates to improve traceability, the development of open-source tools to facilitate interpretation, enhanced validation procedures and quality control and the alignment with FAIR principles through the EM-DAT Archive (CRED Crunch, 2024 and EM-DAT Documentation: Introduction, n.d.).

# Chapter 3 – Data Analysis and Model Comparison

This comprehensive analysis examines machine learning approaches for predicting economic disaster impacts using the EM-DAT database. Recording a 41.5% missingness rate, it represents an interesting opportunity to experiment with techniques that can have a positive impact on the society, trying to bridge the gap and make data complete.

Supervised learning is a machine learning approach where algorithms are trained on labeled datasets containing input-output variables, enabling the models to learn relationships between features for accurate predictions on unseen data. The inputs (features) and the corresponding outputs (labels) are provided during training. Being our label continuous, this research represents a regression task. The models considered in this analysis are Gradient Boosting, XGBoost, Random Forest and Ridge Regression; each complemented with missing data handling techniques, namely Group-wise Median Imputation, Iterative Bayesian Regression and Deletion.

## 3.1 Data Preparation

The implementation begins with data filtering, aimed at excluding data of poor quality and that might contribute to bias perpetration. In line with the guidelines provided by CRED, all pre-2000 events were excluded due to documented reporting inconsistencies, to prevent confusing better performance with actual disaster patterns.

For this thesis, our focus will be only on natural disasters. The rationale behind this choice is the intrinsic objective of finding patterns among existing data related to disasters that would allow to predict future

disasters' impact through damages estimations. Technological disasters, on the other side, often include human actions which are not linked by patterns or relations, hence contradicting the essential goal of this analysis.

Lastly, all records with missing data in the target variable have been deleted. In supervised machine learning, it is essential that every training example includes both the input features and the corresponding target variable because the learning process fundamentally relies on mapping input data to known outputs. Hence records with missing target values do not provide the necessary information to the model to learn from this mapping, as there is no correct answer the algorithm can learn from. In addition, such records do not contribute to the calculation of the loss function, which measures the discrepancy between predicted and actual values and guides the optimization of model parameters. As a result, keeping missing values would possibly introduce inconsistencies and errors in the training process.

### 3.3.1 Feature Engineering

The casualty_impact variable is engineered to capture the nonlinear interaction between mortality and displacement in disasters. The effect of a disaster with 100 deaths and 10,000 affected people is not just the sum of the two, as it creates nonlinear system failures where infrastructure, logistics and human resilience are affected beyond additive limits. The +1 serves as a smoothing factor preventing zero multiplication when deaths occur without affected populations. EM-DAT often has incomplete or inconsistent reporting for "No. Affected" but is more reliable in reporting deaths (Sapir et al., 2015). The ethical principle is that a single death carries different weight in disaster impact assessment than a single affected or displaced person.

```
emdat['casualty_impact'] = emdat['Total Deaths'] * (emdat['No. Affected'] + 1)
```

*Figure 1. Casualty Impact Multiplier*

The 2011 Great East Japan Earthquake triggered a catastrophic tsunami and nuclear meltdown at Fukushima Daiichi, resulting in 19,729 deaths, 2,559 missing persons and 470,000 displaced individuals. Hospitals faced dual pressure, both from people requiring assistance and from patients at risk of death, tsunami waves blocked 561 km of coastal roads and damaged power grids disrupted communication, delaying aid coordination. The interplay of deaths and displacement caused an exponential strain on the country's society, validating the need for a variable that considers the multiplicative, rather than additive, impact of the disaster (Government of Japan, Reconstruction Agency, n.d.).

Disaster impact data exhibits right-skewness, with most events that are relatively minor and a few catastrophic outliers. This skewness could hinder the performance of many machine learning models, especially those that assume normality or are sensitive to extreme values. The logarithmic transformation addresses this skew by computing log(X+1), defined for all X greater than or equal to 0, which compresses the scale of large values and stretches the scale of small values, making the distribution more symmetric and normal.

Optional categorical features, namely Subregion and Associated Types, are added to the list of categorical features only if they are present. This ensures that the feature set is as rich as possible without causing errors if certain columns are missing from the dataset.

Moreover, a crucial point in disaster impact prediction is the heterogeneity of magnitude measurements across different types of disasters.

Earthquakes are measured on the Richter scale (or a moment magnitude scale), tropical cyclones are categorized by wind speed and floods are described by peak discharge. As these units of measure are not comparable, the code first ensures that the Magnitude column is treated as a numerical variable by transforming any non-numeric entry to missing values. Then applies a group-wise percentile rank within each disaster type, ranking each value relative to other events of the same type. Now this value represents the relative intensity of an event within its own hazard category, rather than an absolute value that cannot be compared across types.

This approach has several advantages, among which are the possibility to perform cross-type comparisons and the mitigation of overfitting in disaster types with very few magnitude records. If there are fewer than five non-missing values for a given type, the code assigns a median rank of 0.5; a conservative choice that avoids assigning spurious high or low rankings based on insufficient data. By appending this ranked magnitude to the list of numerical variables used in modeling, the algorithm ensures that information about intensity is incorporated in a normalized and statistically robust manner.

```
for col in ['Subregion', 'Associated Types']:
    if col in emdat.columns:
        categorical_features.append(col)

if 'Magnitude' in emdat.columns:
    emdat['Magnitude'] = pd.to_numeric(emdat['Magnitude'], errors='coerce')
    emdat['magnitude_rank'] = emdat.groupby('Disaster Type')['Magnitude'].transform(
        lambda x: x.rank(pct=True) if x.notna().sum() > 5
        else 0.5
    )
    numerical_features.append('magnitude_rank')
```

*Figure 2. Relative Magnitude*

## 3.2 Choice of Models

### 3.2.1 Missing Data Handling

As the EM-DAT has proven to have multiple missing variables, missing data handling is a fundamental step in data analysis. Two different approaches are taken for categorical and numerical variables.

For categorical variables, the code calculates the frequency of each category. Those that appear fewer than 10 times are grouped together and replaced with the label “Other”, to reduce the impact of rare categories that could otherwise lead to overfitting in the model. Then, any missing value is filled with the label “Unknown”, simplifying the space and preventing issues due to rare or missing categories.

For numerical variables, three methods have been employed, combining adaptability and efficiency for an early-stage analysis.

Row-wise deletion, also known as complete case analysis, involves removing any observation from the dataset if it contains at least one missing value in any of the variables used for the analysis. This method is simple to implement and interpret, as it requires no assumptions about the missing data, and preserves the relationships between data as there is no artificial input (Kang, 2013). However, it can lead to a substantial loss of data and if data are not MCAR, as in our case, and it may introduce bias because the remaining data may not be representative of the full population. Reduced statistical power also plays a role due to smaller sample size.

Median imputation replaces missing values in a variable with the median of the observed values for that variable. In this analysis, this has been further refined by calculating the median within each disaster type. Such method

is resistant to outliers and by stratifying by disaster type, it preserves group-level characteristics and reduces bias compared to global median. Yet, it underestimates the variance of the imputed variable, as all missing values are assigned to the same central value within each group, on top of overlooking relationships between variables. If many values are missing within a group, the imputed value may be unrepresentative, especially in small ones (Zhou et al., 2024).

Bayesian Regression Imputation instead uses probabilistic models to estimate the missing values based on observable relationships between variables. Each variable with missing values is regressed over other variables using this method which incorporates prior distributions and regularizations. max_iter has been set to 20, reflecting Keita's insight on potential model's achievement of 90% gains within 20 iterations in medium-complexity problems (DataCamp, 2024). Random_state has been set to 42 to ensure train and test splits will be identical each time the code is run and ensure reproducibility of the model by controlling sources of randomness during data processing and model training (TowardsDataScience, 2022). This approach allows to preserve the covariance among variables through conditional imputations, can handle MAR data and, if auxiliary variables are included, also MNAR. However, it is computationally intensive, especially for large datasets with many variables with missing data and therefore, if relationships among variables are mis-specified or weak, the imputations may be unreliable (Howey et al., 2017).

Being the data in EM-DAT MAR or possibly MNAR, expectations position Bayesian Regression Imputation as the best method. However, as will be shown by the results of the models, different methods outperformed Bayesian Regression. This outcome could be mainly due to weak correlations between the variables and the absence of auxiliary variables in

the dataset. As visible in these images, no predictive variable showed strong linear correlation with the target variable, suggesting that there may exists non-linear relationships and the variables could still provide complementary information. Moreover, disaster subgroups with few complete cases could provide unreliable covariance estimates, perpetrating bias across the dataset.

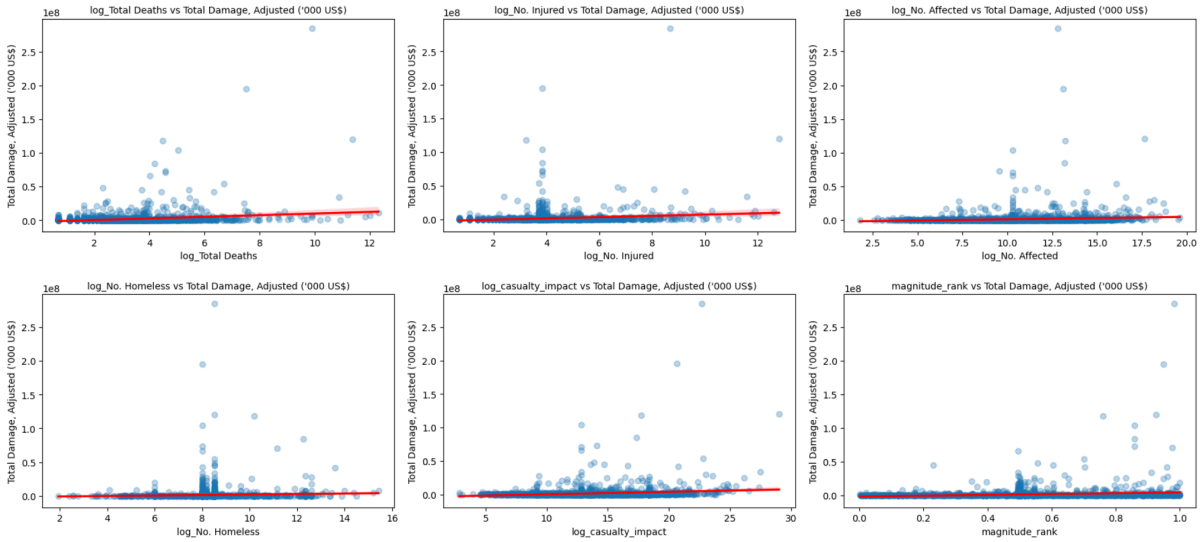



*Figure 3. Scatterplots of some predictive variables vs target variable, Median Imputation*

### 3.2.2 Model Architecture and Hyperparameter Selection

The preprocessing pipeline serves as a solution to critical challenges arising from the EM-DAT's heterogeneous features. The OneHotEncoder with min_frequency=30 and handle_unknown='infrequent_if_exist' contributes to reducing dimensionality by collapsing rare disaster features into infrequent groups and preserving new categories during inference. RobustScaler outperforms StandardScaler by using median and Interquartile range rather than standard deviation to neutralize extreme events skewness, preserving distance between typical events. The train_test_split configuration, set to 0.8 for training the model and 0.2 for testing it, ensures that the model has access to enough examples to learn complex patterns, crucial in datasets where each variable contains valuable

information for generalization. At the same time, it offers a sufficiently large and representative sample to accurately assess the model's performance on unseen data. This ratio is effective with moderate size datasets, as it avoids underfitting resulting from having little training data or vice versa having unreliable variance estimates due to few testing data (Sivakumar et al., 2024).

To effectively predict the economic impact of disasters contained in the EM-DAT dataset, four machine learning models were selected based on their complementary strengths and suitability for the data. Gradient Boosting and XGBoost are powerful ensemble techniques able to capture non-liner relationships and interactions among features, which are common in disaster datasets. Random Forest provides a robust and interpretable baseline that handles categorical variables and outliers well, while Ridge Regression gives a regularized linear perspective, helping to benchmark the performance of more complex models and assess the importance of linear versus nonlinear effects. This diverse set of models enables a comprehensive evaluation of prediction strategies, striving for a balance of accuracy, interpretability and computational efficiency (Jarupunphol et al., 2024).

Gradient boosting works by sequentially constructing decision trees that minimize residual errors though gradient descent. It captures nonlinear relationships through additive modeling and handles heterogenous data types without preprocessing. However, it requires careful tuning, it is prone to overfitting with noisy data without subsampling and its sequential training limits parallelization. For this analysis, n_estimators has been set to 1000, allowing the model to correct errors from previous trees and capture complex nonlinear relationships. The low learning rate of 0.05 improves generalization and reduces overfitting by making a small

adjustment at each tree. However, it requires a higher number of trees to compensate for slower learning (in line with n_estimators=1000). The max_depth of 7 allows to model balance detail and generalization, capturing complex interactions without leading to overfitting for rare or outlier events, while max_features='sqrt' reduces correlation between trees.

XGBoost is an enhanced Gradient Boosting with generalization, implementing sparsity-aware splitting for missing values. It features built-in cross-validation and early stopping, handling large datasets via block structured data layout. In this case n_estimators has been set to 1200 to help capture subtle patterns, the higher learning rate (learning_rate=0.08) allows faster learning, and the deeper trees (max_depth=8) allow to model complex feature interactions, at risk of memorizing noise. However, this model is expected to be the best-performing one thanks to its state-of-the-art performance in many domains (Pakzad et al., 2024).

The Random Forest regressor is an ensemble of decorrelated trees via bootstrap aggregation with features randomization. It is resistant to outliers through majority voting, it embeds implicit feature selection, and it handles multicollinearity. However, it is biased towards categorical variables with many levels, and its memory footprint scales linearly with tree count (Gzar et al., 2022). With lower trees (n_estimators=800) and higher depth (max_depth=15), it aims to capture important and rare patterns while generalizing across different disaster types. Moreover, a split requires at least 8 samples, which prevents the model from creating branches based on rare choices.

Ridge Regression is a linear regression technique that includes L2 regularization, meaning that it adds a penalty proportional to the sum of the squared coefficients of the loss function. This pattern discourages the

model from assigning large weights to any predictor, which is crucial when multicollinearity is present. A standard value of alpha=1.0 helps prevent overfitting without biasing the model, fit_intercept=True ensures that the model learns an intercept term and allows it to adjust the mean of the target variable. max_iter=10000 allows the optimization algorithm enough iterations to converge. However, this model intrinsically assumes linear relationships, which is not our case as shown previously by the scatterplots.

These models offer theoretical trade-offs, with boosting methods reducing bias through sequential correction, Random Forest doing so with bagging and Ridge balancing with L2 constraint.

### 3.3 Model Evaluation and Performance

The selected models have been evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), $R^2$ (Coefficient of Determination) and Adjusted $R^2$. MSE and RMSE have been used for their ability to capture large errors, as outliers are given larger weights. On the contrary MAE is less sensitive to outliers as it does not square the errors, making it robust to outliers. $R^2$ quantifies the proportion of the variance in the dependent variable that is predictable from the independent variables, ranging from 0 to 1 (the best). Adjusted $R^2$ adjusts the $R^2$ value based on the number of predictors in the model, penalizing the addition of unnecessary variables and helping to prevent overfitting.

Among Gradient Boosting, XGBoost, Random Forest and Ridge Regression; each analyzed with three different missing data handling approaches, namely Bayesian Regression Imputation, Median Imputation and Row-wise Deletion; the one who showed higher performance is XGBoost trained with Median Imputation for missing data.

The model achieved $R^2$=0.5865 and Adjusted $R^2$=0.5297, a significant achievement in social science (Ozili, 2022), especially considering the inherent unpredictability of disaster magnitudes and the significant documented data collection inconsistencies. The large values of MSE, RMSE and MAE reflect substantial prediction errors, which are likely due to the wide range and scale of the target variable.

This result is justified by XGBoost's integrated L1/L2 regularization (reg_alpha=0.1, reg_lambda=0.1) which penalizes complex models, reducing overfitting risks while retaining predictive power. The objective function indeed combines loss and regularization terms through number of leaves (count of terminal nodes in a tree) serving as complexity control and leaf weights (prediction values or scores assigned to each leaf) acting as penalties.

XGBoost's tree pruning and max depth=8 effectively capture nonlinear interactions between casualty_impact, magnitude_rank and geospatial features. Column-wise and row-wise subsampling introduces diversity and improves generalization with respect to more static approaches of pother models. Moreover, XGBoost allowes for domain-specific adaptations that better help describe the EM-DAT. Its TransformedTargetRegressor(log1p) mitigates skewness, while regularization and feature subsampling prevent overfitting to rare categories.

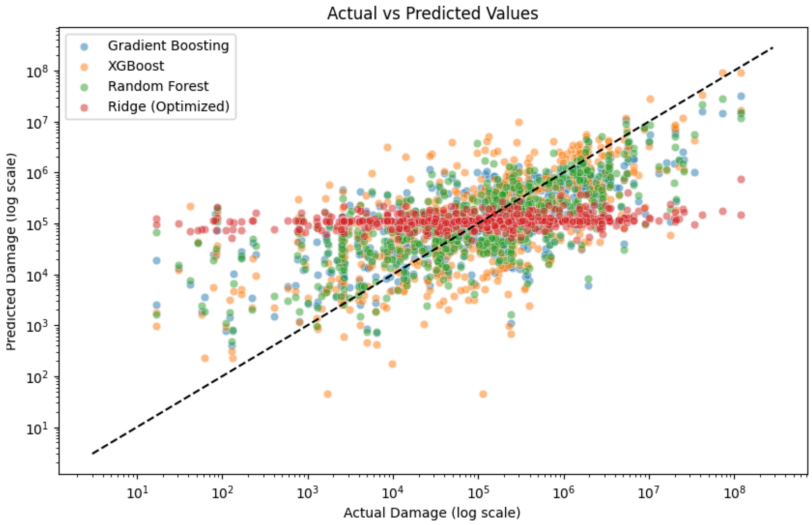



Figure 4. Actual vs Predicted Values, Median Imputation

Feature importance analysis has been used to reveal key predictors across models. The prominence of casualty impact validates the feature engineering approach and confirms the nonlinear relationships between human casualties and economic damage. The presence of geographic features also highlights possible regional variations in reporting practices, reflecting known biases in the EM-DAT dataset.

### 3.3.1 Economic Impact Ratio

The model also includes another engineered feature, called Economic Impact Ratio (EIR). This ratio normalizes damage estimates by human impact, proving insight into the economic-to-human impact relationship.

```
predictions_df['Actual_Economic_Impact_Ratio'] = (
    predictions_df['Actual_Damage'] /
    (predictions_df['casualty_impact'] + 1)
)
```

*Figure 5. Economic Impact Ratio*

The formula includes the addition of 1 to the denominator, to prevent division by zero and consequent errors for events with no casualties. The ratio reflects that economic damage and human casualties are different dimensions of disaster severity, yet equally relevant and consistent with

social accounting matrix frameworks (Botzen et al., 2019). The selection of an EIR threshold of 2.0 is grounded in empirical disaster economics literature and policy frameworks for fiscal response optimization and serves as a critical benchmark to identify when economic damage disproportionately outweighs human casualties. Empirically, it captures the top quartile of disasters in terms of economic damage relative to human impact, aligning with OECD findings that 20-25% of events drive 80% of cumulative losses (OECD, 2018). Specifically, it corresponds to the IMF's identified inflection point of 7.1% damage-to-GDP, beyond which macroeconomic stability is significantly compromised (Nguyen et al., 2025). It provides a replicable decision boundary for activating contingency financing instruments, infrastructure protocols and international aid requests, positioning itself as an innovative advancement given that 63% of countries lack standardized disaster impact thresholds (Hallegatte et al., 2024). While many ratios operate in GDP terms, the EIR metric normalizes the logic by measuring economic damage relative to human impact, an approach validated by World Bank's loss/exposure ratio.

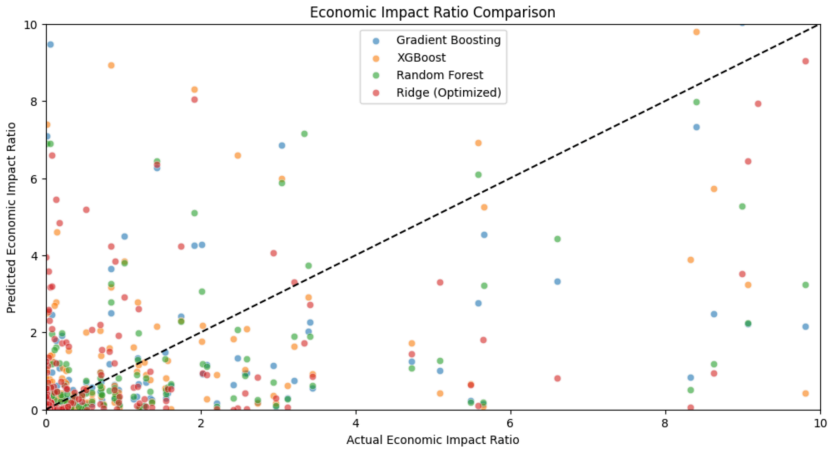



*Figure 6. Actual vs Predicted for Economic Impact Ratio, Median Imputation*

When analyzing Actual vs Predicted values for EIR, under Median Imputation hypothesis, we note that the model's systematic

underprediction (13.67-14.63%, except Ridge Regression with 16.4%, vs 15.67% actual) suggests incomplete capture of multi-year reconstruction dynamics to amplify economic impacts, supply chain cascade effects where infrastructure damage creates non-proportional losses and price inflation effects. XGBoost's balanced performance (14.63%) demonstrates how L1/L2 regularization prevents overfitting to localized disaster patterns, while linear models struggle with thresholds discontinuities. Moreover, tree ensembles better approximate the capital stock recovery functions observed in macroeconomic models (Hallegatte et al., 2022).

# Chapter 4 – Ethical Discussion

## 4.1 CapAI Framework Overview

The Conformity Assessment Procedure for AI (capAI) framework represents a groundbreaking approach to AI systems conformity assessment, designed specifically to align with the EU Artificial Intelligence Act requirements. It was developed by a team of researchers led by the University of Oxford and the University of Bologna, with key contribution of Professors Luciano Floridi and Matthias Holweg supported by the Oxford Internet Institute and the Centre for Digital Ethics at the University of Bologna (Oxford Internet Institute, 2022). The framework addresses the critical gap between high-level ethical principles and practical implementation guidance for AI developers and deployers.

The conformity assessment procedure encompasses three core components that work synergistically to provide a holistic evaluation of systems. The Internal Review Protocol (IRP) is the primary governance tool that offers organizations a structured approach to AI quality assurance and risk management. It follows the AI system's development stages and assesses organizational awareness, performance and resources put in place to prevent and address eventual failures. The Summary Datasheets (SDS) are standardized documentation that facilitates transparency and stakeholder communication. Lastly, the External Scorecard (ESC) is an optional tool that provides stakeholders with an overview of the system's health status regarding good practices and ethical systems adherence (Floridi et al., 2022).

The capAI methodology adopts a process-oriented view of AI systems, recognizing the dynamic nature of machine learning systems and the need

for adaptive governance structures that can respond to unpredicted risks. The framework covers five critical stages of AI lifecycle: design, development, evaluation, operation and retirement. Each stage presents unique challenges and opportunities for ensuring ethical AI implementation. Central to the capAI approach is the concept of trustworthy AI, which encompasses three essential requirements that AI systems must fulfil throughout their entire lifecycle. Systems must be legally compliant, adhering to all applicable laws and regulations; ethically sound, following ethical principles and values; and technically robust, maintaining reliability from both technical and social perspectives.

### 4.2 Algorithm Conformity Assessment

In a context such as disaster management, technical evaluation is not sufficient to determine an algorithm's success. Having more stakes than the business ones, following EU AI regulations represents a must for any proposed implementation, which serves as a guarantee of protection of social interest. Similarly, the following ethical audit aims to evaluate the model proposed previously considering applicable regulations, following the capAI framework proposed by Floridi and Holweg.

#### 4.2.1 Legal Compliance

The EM-DAT aggregates disaster data from more than 40 sources, including UN agencies, national governments and NGOs, providing multiple validation points that enhance data reliability and reduce the risk of systematic bias. The algorithm therefore inherits EM-DAT's privacy protection through aggregated reporting formats, eliminating personal data risks. The database's open access availability for non-commercial use aligns well with transparency principles embedded in the capAI framework. Lastly, the complete explanation of feature engineering, model selection and data

preprocessing are crucial for ensuring that the system meets the explicability requirements outlined in the EU AI guidelines.

As a disaster prediction tool affecting emergency resource allocation, the system falls under AI Act (Regulation (EU) 2024/1689, AIA) Annex III high-risk classification, requiring mandatory conformity assessments, human oversight protocol and real-time performance monitoring. As of this initial version, the system does not show any prohibited practices as per Article 5 of AIA (EU AIA, 2025).

### 4.2.2 Ethical Soundness

The capAI framework highlights four core ethical principles that are crucial for any AI system: respect for human autonomy, prevention of harm, fairness and explicability. The current implementation positions the algorithm as a decision-support tool rather than an automated decision-maker, which aligns with capAI's human autonomy principle. However, in case of deployment there should be user training programs for interpretation, given the high uncertainty due to the biases documented in the dataset and the model uncertainty metrics. The prevention of harm principle requires comprehensive risk assessment and mitigation strategies throughout the AI lifecycle. Disaster-related algorithms carry inherent risks as "false positives" which in this case relate to over-allocation of resources, and "false negatives", in this case under-allocation. Moreover, the EM-DAT's inclusion criteria create inherit sampling bias, with underrepresentation of small-scale disasters in developing countries. A good future practice would be the implementation of subgroup analysis using EM-DAT's disaster subtypes with SHAP value reporting, to examine whether the algorithm produces equitable outcomes across different populations, geographic regions and disaster types.

### 4.2.3 Technical Robustness

The framework requires comprehensive documentation of model validation approaches, to examine whether the validation approach accurately captures the algorithm's expected operating environment. The algorithm uses post 2000s temporal range for cross validation, but due to this lacking in the original dataset, does not account for climate change adjustment factors. However, it inherits EM-DAT's protection against adversarial attacks through data aggregation protocols and multi-layered access controls (EM-DAT Documentation, n.d.).

### 4.2.4 Transparency and Accountability

The capAI framework places significant emphasis on transparency and accountability mechanisms that enable stakeholders to understand and evaluate AI system behavior. In the case of disaster-related applications, transparency requirements include documentation of the algorithm's intended use cases and limitations. Accountability instead refers to clear definition of responsibilities for algorithm development, deployment and ongoing monitoring, which, being the algorithm an elementary version, are not in place yet. Lastly, traceability requirements are satisfied through comprehensive documentation of the algorithm's development process, including data sources, model architecture decisions and validation approaches.

## 4.3 Concerns

The application of machine learning in disaster management raises several ethical concerns that must be carefully addressed to ensure responsible and effective implementation (Gevaert et al., 2021).

### 4.3.1 Bias and Fairness

Machine learning models may inherit and amplify existing bias present in the training dataset, therefore leading to unfair or discriminatory outcomes. In the context of disaster management, this translates to inequitable resource allocation or response prioritization. To address these concerns, explained in detail in Section 2.3, there should be regular third-party audits of the EM-DAT dataset and models built with its data to implement fairness constraints and bias mitigation techniques.

### 4.3.2 Privacy and Security

While it is true that the EM-DAT dataset does not contain personally identifiable information, privacy and security remain relevant for the broader context. Security implications should be evaluated when deploying the solutions with real-time data that may contain sensitive information, respecting the privacy of affected communities and individuals.

### 4.3.3 Transparency and Explainability

Machine learning models often function as "black boxes" that make predictions without clear explanations of their reasoning. In disaster management, where decisions have life-or-death implications, this lack of transparency is problematic. For this purpose, the research has included comprehensive documentation of model development, including data sources, preprocessing steps, feature selection and evaluation methods.

### 4.3.4 Hype and Misinterpretation

There is significant risk of overreliance on the models' predictions, potentially leading to unrealistic expectations. For this reason, the models are intended to serve only as a support tool in decision making, which should not in any case substitute human judgement. Future applications

should therefore include resources to help practitioners understand both the capabilities and limitations of the developed models.

### 4.4 Principles for Digital Development

The development and deployment of machine learning models for disaster impact assessment must adhere to established principles for responsible digital development. The Principles for Digital Development provide a valuable benchmark to ensure that technological solutions address real-world needs effectively and ethically. The principles have originally been developed in 2014 and endorsed by more than 300 organizations. After being updated in 2024, they now better reflect nowadays' larger interactions between people and technology. They recognize the need for radical inclusion and local ownership, elevated issues arising from the generation and use or digital data, emphasized open approaches to innovation and resonate with the diversity of individuals and organizations holding power over the AI lifecycle. When designing and implementing a solution, developers adhering to the principles should at minimum not harm, and at best ensure their work maximizes the agency of people involved to drive the development (DigitalPrinciples, n.d.).

#### 4.4.1 Alignment with Principles for Digital Development

Understanding the Existing Ecosystem: The research acknowledges the complex landscape of disaster management, including already existing national frameworks and the roles of individual stakeholders. The machine learning models have been designed to complement, rather than substitute, the existing frameworks.

Share, Reuse and Improve: The research uses a dataset open to the public for non-commercial use. The models, code and findings have been

documented in the previous chapters and are publicly available through this research.

Design with People: The models have been developed keeping in mind that their ultimate objective is to serve the needs of affected communities, hence stakeholders' perspectives should and will be considered for further refinement.

Design for Inclusion: Due attention has been given to ensuring the models do not perpetuate or exacerbate existing inequalities. This includes detailed documentation and management of potential biases in EM-DAT.

Build for Sustainability: The proposed models have been selected also because of their ease of use and development, aiming at guaranteeing their long-term sustainability in case of changing conditions and data availability.

Establish People-First Data Practices: The research has clearly stated limitations and biases to ensure ethical deployment of the models and appropriate uses.

Create Open and Transparent Practices: The development process is transparent, with clear and available documentation of methodologies, assumptions and limitations. Such process is fundamental to build trust in the models and enable appropriate output interpretation.

Anticipate and Mitigate Harms: Potential negative consequences of model deployment have been identified and will be addressed.

Use Evidence to Improve Outcomes: The research should continuously evaluate model performance and impact though empirical evidence, to ensure a positive contribution to disaster management practices.

By adhering to these principles, this research aims to develop solutions that are not only technically sound, but also ethically responsible and able to address real-world challenges responsibly.

### 4.5 Ethical Evaluation

The capAI conformity assessment reveals that the algorithm has strong foundational alignment with both AI governance principles and digital development objectives, while some areas still need significant improvement. The use of the globally recognized EM-DAT dataset positions it well within established risk reduction frameworks and demonstrates commitment to transparency (IBRD, 2016). It supports the broader objective of making disaster risk information accessible to diverse stakeholder communities, with further action to be taken to extend implementation beyond technical aspects. Moving forward, the successful implementation of capAI recommendations will require sustained commitment to documentation enhancement, governance structure development and stakeholder engagement process, to support the algorithm's evolution from a technical tool to a trusted component of disaster management ecosystems. The investment in these governance mechanisms seeks to ultimately strengthen the algorithm's social acceptance, contributing to the broader objective of building trustworthy AI systems that serve the public good while respecting ethical principles and legal requirements.

## Chapter 5 – Conclusion

This thesis demonstrates the potential that Machine Learning has in enhancing economic impact prediction in disaster management but also highlights that it requires careful navigation of data limitations and ethical complexities.

The XGBoost model with Median Imputation emerged as the most effective predictor ($R^2$=0.5865), validating the importance of engineered features like the casualty impact multiplier in capturing nonlinear disaster dynamics. However, EM-DAT's systemic biases, including 96.2% missing reconstruction costs and geographic underrepresentation in low-income regions, highlights inherent constraints in data quality that require cautious interpretation of model outputs.

The ethical analysis using the capAI framework revealed gaps in alignment with EU AI regulations, particularly in bias mitigation and human oversight control. While the models follow practices under Article 5 of the AI Act, their high-risk classification mandates rigorous conformity assessments and real-time monitoring in operational settings. Adherence to the Principles for Digital Development emphasizes stakeholder inclusion, open-source collaboration and safeguards against algorithmic inequality.

Future key recommendations include the integration of SHAP analysis to audit subgroup fairness, expanding EM-DAT's data validation protocols and developing hybrid models that directly include the community. Future research should also include climate change-adjusted validation frameworks. By prioritizing both technical robustness and ethical governance, this thesis advances the discourse of AI for social good, offering actionable insights for policymakers and practitioners committed to fair disaster resilience.

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